

dimensions. The Hooke-Jeeves method is susceptible to local minima. The method has been proven to converge on certain classes of functions.⁴

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function hooke_jeeves(f, x,  $\alpha$ ,  $\epsilon$ ,  $\gamma=0.5$ )
    y, n = f(x), length(x)
    while  $\alpha > \epsilon$ 
        improved = false
        best = (x=x, y=y)
        for i in 1 : n
            for sgn in (-1,1)
                x' = x + sgn* $\alpha$ *basis(i, n)
                y' = f(x')
                if y' < best.y
                    best, improved = (x=x', y=y'), true
                end
            end
        end
        x, y = best
        if !improved
             $\alpha *= \gamma$ 
        end
    end
    return x
end
```

⁴ E. D. Dolan, R. Torczon, "On the convergence of Pattern Search", *Journal on Optimization*, no. 2, pp. 567–58

Algorithm 7.5. The Hooke-Jeeves method, which takes as input a function f , a starting point x , an initial step size α , a step decay γ . The algorithm iterates until the step size is less than ϵ . The points sampled in the coordinate directions are used to determine improvement. Based on Algorithm 178: Direct Search, *Communications of the ACM*, pp. 313–314, 1965.